

Scalability & Future Plan

Date	2 October 2026
Team ID	12
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	One request blocks until all images are done (about 104 s)	Poor experience, ties up the server	High
2	Gemini free tier: 20 requests/day per model	About 20 comics per day per model pair	High
3	No character consistency; no speech bubbles; CJK text unsupported in PDF	Lower comic quality	Medium

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single local process, one comic at a time	Job queue with GPU workers; return a job id and poll for progress
2	Data Storage	PNG and PDF files on local disk	Object storage (e.g., S3) with clean-up of old files; optional database for saved comics
3	Performance	fp16 on MPS, warm-up, parallel story	Cloud GPU, batched or faster models, cache outputs per prompt
4	Security	API key in .env, no user data	Add login and rate limiting before public hosting; paid Gemini tier or google-genai SDK

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Progress UI per panel, styled preview and export pages, CI running the pytest suite	1 month	Better experience; safer changes

Phase 3	Speech-bubble overlay and character consistency (IP-Adapter or LoRA)	3 months	Higher comic quality
Phase 4	Docker deployment on a cloud GPU, accounts and gallery, multi-language support	6 months	Public availability and wider audience